

Geocoded Analysis of the Determinants of Socio-Political Unrest in South Africa

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Abstract:

Protests have severe negative short and long-run impacts on the developing South African economy. They have caused significant increases in costs and losses in productivity for crucial South African industries, leading to declining growth rates and further increases in unemployment and poverty.

The focus of this study was to examine how service delivery, income level, and political satisfaction affect the intensity of unrest in South Africa. Secondary geocoded data from the GDELT project and Afrobarometer was utilised to construct composite indices to analyse the relationships between unrest intensity and its predictors.

In line with the predictions of Frustration-Aggression Theory and studies conducted by Kali (2023), Nleya (2011), and Botes (2018), the study revealed a positive linear relationship between poor service delivery and unrest intensity.

The findings contribute to a better understanding of the complex dynamics between unrest and its determinants. Policymakers and stakeholders can leverage this understanding to address underlying issues effectively and work towards mitigating and reducing socio-political unrest. The study's framework, utilizing GDELT data, can also be applied to investigate unrest in other countries, benefiting from the broad global coverage provided by GDELT.

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1. Introduction

According to Kali (2023), the root causes of unrest in South Africa can largely be ascribed to three factors, namely; socio-economic situations, quality and reliability of service delivery, and political satisfaction. There can, however, be difficulty in identifying the exact causes that triggered unrest due to the level of unrest often not being proportional to the extent of the reported cause and frustrations not being directly linked to the cause. During protests and demonstrations, there are often also participants without a direct stake. To address this issue, it is necessary to analyse and understand the root causes of frustration among protest participants. By identifying and addressing these underlying concerns, efforts can be made to prevent or minimize the frequency of protests in the first place.

Shonchoy and Tsubota (2015) examine how protests can have a devastating effect on economic activities and often lead to damage to both private and public infrastructure. This destruction of property and disruptions in work lead to significantly increased costs for businesses as well as a drastic reduction in productivity.

Tenza (2020) expounds on these effects further with a specific focus on the developing South African economy. Due to these significant increases in costs and loss of profits from long periods of non-production, businesses often retrench their workers, leading to a further worsening of poverty levels in South African communities. Protests and unrest also hamper the creation of employment opportunities, yet again leading to a further worsening of poverty levels, a reduction in the potential for future economic growth, and an increase in future unemployment rates as population growth rates exceed the country's ability to create employment opportunities. The South African government also heavily relies on income tax as a crucial source of revenue with growing unemployment, due to protests and unrest, being an important eroding factor of the country's income base. The severe negative effects described by Tenza (2020) could be observed during the violent and prolonged mining sector strikes from 2012, the mining sector reported billions of Rands in losses and decreases in output, leading to a drastic slowdown in economic development, which was already on a downward scale at the time.¹

With all the cascading factors taken into account, a vicious cycle is created that can have current and long-term negative impacts on the economic well-being of the country. It is thus important to devise an effective method that can investigate to what degree potential predictors, such as poverty, poor service delivery, and political dissatisfaction, affect protest intensity.

¹ <https://www.news24.com/news24/mining-industry-lost-billions-during-strikes-20150429>

The GDELT project captures global occurrences, assesses sentiment associated with reporting, and facilitates automated analysis on a large scale. It is considered reliable due to continuous content updates and its extensive global coverage (Saz-Carranza, Maturana, & Quer, 2018). GDELT has thus become a valuable resource for many academic researchers and is used in this study.

Geolocated data from the GDELT project as well as Afrobarometer was utilised to construct markers of social grievances that were used to predict probabilities of unrest. The study found that there is a positive linear relationship between the intensity of unrest and poor service delivery. In areas with worse service delivery, there is thus a higher probability of experiencing increased protest intensity.

The findings of this study help improve our understanding of the relationships between unrest and its predictors. The framework employed in this research can also be utilised to investigate unrest in other countries, leveraging the global coverage offered by GDELT.

2. Review of Literature:

Negative Impacts of Protests and Unrest

Protests and unrest have significant negative implications for the economy of a country. Research by Barrett and Chen (2021) reveals that protests result in a reduction in GDP, with the impact being more severe if the protests are motivated by socioeconomic factors rather than political ones. Furthermore, protests motivated by both factors lead to even greater contractions in GDP. The effects of protests go beyond economic growth; they also cause damage to private and public infrastructure, resulting in increased costs for businesses and a decline in productivity (Shonchoy & Tsubota, 2015).

Tenza (2020) focuses specifically on the impact of protests on the developing South African economy. Prolonged periods of non-production and rising costs compel businesses to lay off workers, exacerbating poverty levels within South African communities. This creates a detrimental cycle of unrest and protests. Additionally, protests and unrest hinder the creation of employment opportunities, further worsening poverty levels, reducing the potential for future economic growth, and increasing future unemployment rates as population growth outpaces job creation.

Positive Impacts of Protests and Unrest

Protests and social unrest in Africa have the potential to bring about positive social change by drawing attention to various pressing issues. These include the lack of transparency in electoral processes, oppressive policies implemented by governments, dictatorships, and the undemocratic extension of presidential term limits. These demonstrations serve as an avenue for people to express their concerns and advocate for improved governance (ShareAmerica, 2014).

By taking part in protests, citizens can effectively highlight the need for reforms, increased accountability, and the upholding of democratic principles. Through their collective actions, they send a clear message to those in power that change is demanded and expected. Consequently, these social movements play a significant role in shaping a more inclusive and just society. They create a platform for marginalized voices to be heard and contribute to the formation of policies and practices that better address the needs and aspirations of the population at large (Maganga, 2020).

Frustration-Aggression Theory

Frustration-Aggression Theory, or the frustration-aggression hypothesis, suggests that aggression is a response to frustration. Dollard et al. (1939) initially proposed this theory, which has been widely studied and applied in various fields such as ethnology, criminology, sociology, medical research, and psychology. According to the theory, aggressive behaviour emerges when there is sufficient frustration.

In the context of this theory, frustration is defined as an interference with the occurrence of a desired goal-response. It is considered an event rather than an emotional state. Defining frustration based on observable events allows for testing and describing potential causal effects without relying solely on subjective self-reported introspection. A later revision of the theory argued that while frustration may not always lead directly to aggressive behaviour, it will generate instigations for various responses, which could include aggression (Miller, Mowrer, Doob, Dollard, & Sears, 1958).

Morlan (1949) further suggests that although expressing frustration through aggressive impulses might provide a form of catharsis, it does not always dissipate the individual's frustration. The effectiveness of catharsis depends on the individual's attitude towards their frustration. If frustration persists after an aggressive response, it can create a vicious cycle that leads to further aggression and unrest. When examining unrest in South Africa, frustration may stem from widespread poverty, inequality, poor service delivery, inadequately implemented government policies, and political motivations.

Successful Protests and Overcoming the Vicious Cycle

Ten elements of a successful protest movement were identified by De Haan et al. (2020), dialogue as the priority, a clear communication of wants, and heeding all concerns were identified as the most important factors. If these elements are followed, the chances of a successful protest leading to positive social change increase significantly. Moreover, it is unlikely that positive protest outcomes resulting from these elements will contribute to a cycle of escalating frustrations and further protests.

The Bandwagon Effect

According to Karklins and Petersen (1993), the decision to participate or refrain from participating in collective action is often influenced by the number of citizens already involved in the protest. This phenomenon is known as the bandwagon effect, where individuals are more likely to join a protest when they see others participating. The bandwagon effect can help explain the involvement of individuals who may not have a direct stake in the protest. The more participants there are, the lower the perceived risks of individual involvement (Gavious & Mizrahi, 2001).

Causes of Protests in South Africa

Studies on the causes of unrest in South Africa share similar themes. They point to rising frustrations due to poor socio-economic conditions, poor service delivery, or dissatisfaction with governmental response to deal with these issues as the root causes of protests in South Africa. Kali (2023), Nleya (2011), Botes (2018), and Ndasana et al. (2022) link poor service delivery and poverty as major contributors to protest participation in South Africa. These root causes align well with the predictions of Frustration-Aggression Theory, if these problems persist, lingering frustrations are cultivated that eventually lead to aggressive outbursts in the form of protests. The studies do, however, not make use of econometric techniques to analyse to what extent these factors contribute to protest intensity. The current study builds on the existing research by using such techniques for a robust analysis.

3. Research Methodology

Data Collection Methods

The main focus of this research project was on quantitative research, utilizing secondary research data sourced from both Afrobarometer and the GDELT project. To examine the landscape of protests in South Africa, geolocated protest data pertaining to the period of 2018 to 2022 was collected from the GDELT project (event code 14). Concurrently, geolocated survey data in Likert scales were obtained from Afrobarometer, encompassing rounds 7 to 9, allowing for a comprehensive analysis of the given period. The GDELT data consisted of 77 952 entries while the Afrobarometer data consisted of 5020 entries after rounds 7-9 datasets were processed and merged into a new dataset. Single-point Haversine distances were calculated to map longitude and latitude values to South Africa as a cursory examination of their intersection.²

Table 2 shows the count of Afrobarometer data points per province. Gauteng, KwaZulu-Natal, Western Cape, and the Eastern Cape contained the most data points.³

Table 1: Afrobarometer Data Points Per Province

Province	Count	Province	Count
Eastern Cape	583	Mpumalanga	363
Free State	313	North West	334
Gauteng	1221	Northern Cape	303
KwaZulu-Natal	838	Western Cape	609
Limpopo	452		

Table 3 shows the count of recorded GDELT protests observed per province. Gauteng, Western Cape, Free State, and KwaZulu-Natal had the highest recorded number of protests.⁴

Table 2: GDELT Protests Per Province

Province	Count	Province	Count
Eastern Cape	5759	Mpumalanga	971
Free State	9633	North West	3154
Gauteng	31821	Northern Cape	4032
KwaZulu-Natal	8954	Western Cape	11858
Limpopo	1674		

² See Figure 1 for distribution of data

³ See Table 2: Afrobarometer Data Points Per Province

⁴ See Table 3: GDELT Protests Per Province and Figure 2: Protest Intensity Per Province

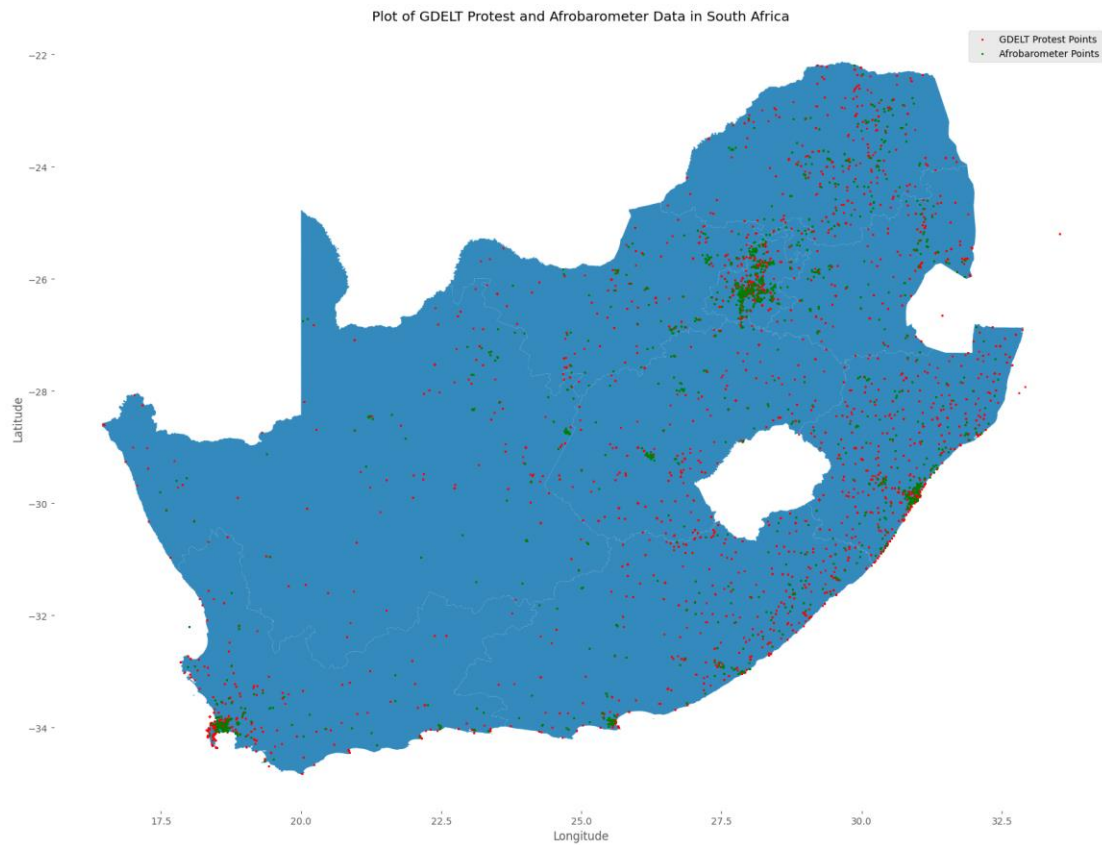


Figure 1: Plot of GDELT Protest and Afrobarometer Data in South Africa

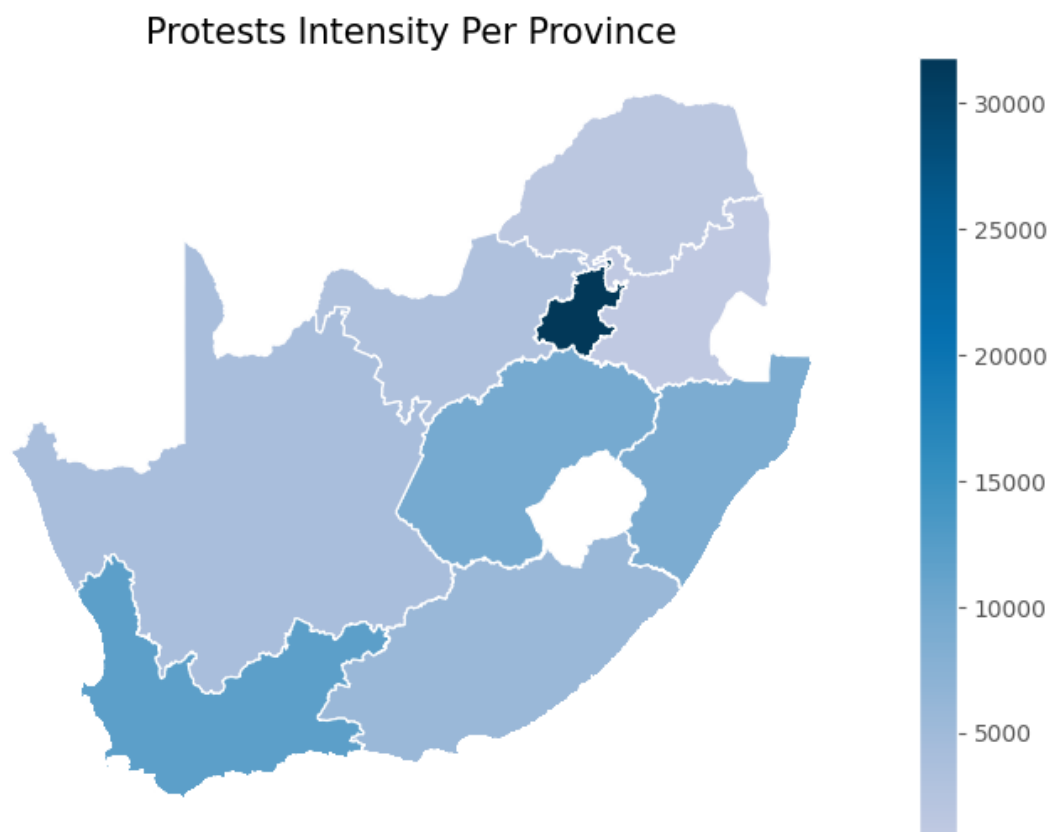


Figure 2: Protest Intensity Per Province

Index Construction

Survey questions from rounds 7-9 of the Afrobarometer surveys were divided into three categories, namely; questions related to levels of income/poverty, questions related to levels of service delivery, and questions related to levels of political satisfaction.⁵ These three categories were selected as protest intensity predictors due to their relationship with protests as established by Kali (2023), Nleya (2011), Ndasana et al. (2022), and Botes (2018).

While questions were labelled differently between the rounds, round 7 questions were used as a base with the equivalent questions from rounds 8 and 9 being renamed and grouped. Table 1 shows which questions were selected from the Afrobarometer surveys and how they were categorised.⁶

Table 3: Afrobarometer Questions

Income Questions	Renamed	Service Delivery Questions	Renamed	Political Satisfaction Questions	Renamed
EA_FAC_B	WalkSchool	EA_ROAD_C	RoadCondition	Q3	CountryDirection
EA_FAC_C	WalkPolice	Q49B	SchoolService	Q4A	CountryEconomic
EA_FAC_D	WalkClinic	Q49E	ClinicService	Q36	Condition
Q4B	LivingCondition	Q49P	PoliceService		Democracy
Q8A	NoFood				
Q8B	NoCleanWater				
Q8C	NoMedicine				
Q8D	NoFuel				
Q8E	NoIncome				

The three categories were encoded using ordinal encoding or encoded as binary variables. Higher values were assigned to responses more indicative of low income, poor service delivery, and political dissatisfaction. The merged and encoded questions were then used to construct three composite proxy indices that were used to analyse how low income, poor service delivery, and political dissatisfaction contribute to the prevalence of protest intensity.⁷ The following indices were thus created; 'Income Level Proxy Index', 'Service Delivery Proxy Index', and 'Satisfaction Proxy Index'.

⁵ See Table 1: Afrobarometer Questions to see what questions were selected and how they were divided into categories

⁶ See Appendix for Afrobarometer survey questions used and how they were encoded

⁷ See equation 1-3

Income Level Proxy Index

$$= \text{WalkSchool} + \text{WalkPolice} + \text{WalkClinic} + \text{LivingCondition} + \text{NoFood} + \text{NoCleanWater} + \text{NoMedicine} + \text{NoFuel} + \text{NoIncome} \quad (1)$$

Service Delivery Proxy Index

$$= \text{RoadCondition} + \text{SchoolService} + \text{ClinicService} + \text{PoliceService} \quad (2)$$

Satisfaction Proxy Index

$$= \text{CountryDirection} + \text{CountryEconomicCondition} + \text{Democracy} \quad (3)$$

where WalkSchool, WalkPolice, WalkClinic, and CountryDirection are binary variables.⁸ While LivingCondition, NoFood, NoCleanWater, NoMedicine, NoFuel, NoIncome, CountryEconomicCondition, and Democracy are ordinal variables.

Protest/Unrest Variable Construction:

The Haversine formula as well as the Spherical Law of Cosines were used to calculate the distances between each Afrobarometer data point and the GDELT recorded instances of protests, making use of latitude and longitude coordinate pairs contained in both datasets. Afrobarometer household data was geospatially linked with the GDELT records that occurred within a radius of 25km.

For a recorded instance of protests to be considered relevant to an Afrobarometer data point, two requirements had to be fulfilled; the protest had to occur within a 25km radius of the data point, and the protest had to occur within the same year or the following year of the data point. A 25km radius was selected as a reasonable distance to indicate a data point's relevant vicinity of available services and employment while a two-year period was used for each data point to attempt to capture present lingering frustrations caused by a continuous state of poor socio-economic conditions and dissatisfaction, as laid out by Frustration-Aggression Theory. If frustrations due to the socio-economic conditions of data points started to build up in a given year, there would be a potential increase in protest intensity in the following year. It was thus deemed important to attempt to capture this effect.

The variable, 'Number of Protests in 25km radius', indicating protest intensity within a 25km radius that occurred within the given year as well as the following year was thus created.

⁸ I am aware that adding binary variables to ordinal variables will underweight them, more complex indices with weights attached to them will be left for future work

Estimation Strategy:

After the Afrobarometer and GDELT datasets were mapped and geospatially joined and indices were constructed, OLS regression was used to analyse protest intensity due to a hypothesised linear relationship between the constructed indices and protest intensity. Province-level fixed effects were used to account for unobserved heterogeneity across provinces.

The following estimation was thus set up:

$$\begin{aligned} & \text{Number of Protests in a 25km radius} \\ &= B_0 + B_1 \text{Income Level Proxy Index} + B_2 \text{Service Delivery Proxy Index} \\ &+ B_3 \text{Satisfaction Proxy Index} + B_4 \text{Province} + \varepsilon \end{aligned} \quad (4)$$

where the Province variable was the province-level fixed effect that was divided into each of South Africa's nine provinces by making use of one-hot encoding. Eastern Cape was dropped to avoid the dummy variable trap.

4. Results and Discussion

Table 4 shows the regression results obtained from the OLS regressions. Regression 1 contained the constructed composite indices but did not include the location fixed effect at province level.⁹ Regression 2 was selected as the final model, containing both the constructed composite indices as well as the location fixed effect at province level.¹⁰ The variable 'Service Delivery Proxy Index' was omitted in Regression 3 to analyse how the relationships between the variables changed if it was not included.¹¹

Table 4: Regression Results

	Regression 1	Regression 2	Regression 3
Income Level Proxy Index	-23.6901*** (3.453)	-4.8255* (2.824)	-2.9577 (2.813)
Service Delivery Proxy Index	33.8836*** (6.203)	28.2335*** (5.006)	
Satisfaction Proxy Index	8.2653 (9.205)	-9.7045 (7.477)	-8.3705 (7.496)
Constant	724.1701*** (69.003)	24.4345 (70.593)	135.2193** (68.013)
N	5020	5020	5020
Adj. R-sq	0.013	0.361	0.357

Standard errors in parentheses.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.010$.

⁹ See Table 4: Regression Results, Regression 1

¹⁰ See Table 4: Regression Results, Regression 2

¹¹ See Table 4: Regression Results, Regression 3

Significant Features:

The variable Service Delivery Proxy Index was found to be highly statistically significant at a 99% confidence interval, indicating a significant positive linear relationship. It is thus evident that poor service delivery greatly contributes to the intensity of protests in a 25km radius in any given year as well as the following year.

The variable Income Level Proxy Index was found to be marginally statistically significant at a 90% confidence interval, indicating a marginal negative linear relationship. This was an unexpected result, consequently, the regression was run again without Service Delivery Proxy Index to analyse the relationships with the variable being omitted. Neither Income Level Proxy Index nor Satisfaction Proxy Index were even marginally significant after Service Delivery Proxy Index was removed and will thus not be considered.¹²

In line with the findings of Kali (2023), Nleya (2011), and Botes (2018) and the predictions of Frustration-Aggression Theory, poor service delivery plays a major role in protest intensity. Continued poor service delivery cultivates lingering frustrations that eventually build up to aggressive outbursts such as protests.

The potential reasoning behind the lack of significance observed in the relationships between certain variables will be briefly examined. One possible explanation for the absence of significance between income level and the intensity of protests within a 25km radius in a given year, as well as the subsequent year, could stem from a pervasive apathy among the population towards economic conditions. Additionally, there might be a high opportunity cost associated with participating in protests, which reduces the likelihood of individuals mobilizing. A more complex index with attached weightings could lead to better results in understanding how economic hardships contribute to protest intensity.

Another explanation for the lack of significance between dissatisfaction in the country and protests could be attributed to a negative association between protests and this specific reason for dissatisfaction. It is plausible that protests are not widely perceived as an acceptable means of expressing discontent in response to general political dissatisfaction. Alternatively, there may exist a generalized belief among the population that protesting is unlikely to resolve either economic or political dissatisfaction-related issues, while the potential for protesting to address poor service delivery might be seen as more promising and more likely to lead to change.

¹² See Table 4: Regression Results, Regression 3

These potential explanations could not be checked from the data used in the current study but might offer interesting avenues for future research.

5. Conclusion and Future Work

Conclusion

This study aimed to examine the relationships between the predictors of unrest and the intensity of unrest within the context of South Africa. Its primary focus was to assess the extent to which each predictor contributes to the observed intensity of unrest. The findings revealed significant associations between protests occurring within a 25km radius of Afrobarometer data points in a given year and the subsequent year if the quality of service delivery in the area was poor.

The geocoded data from Afrobarometer was utilized to construct composite indices. These indices were created by categorizing and encoding Likert-scale survey questions, with higher values indicating higher levels of poverty, poor service delivery, or political dissatisfaction, respectively. These composite indices served as proxies for the predictors of unrest. On the other hand, geocoded protest data was collected from the GDELT Project.

After the Afrobarometer and GDELT datasets were mapped and geospatially joined and indices were constructed Ordinary Least Squares (OLS) regression was used to investigate the relationship between the constructed indices and protest intensity. A location fixed effect at the province level was included in the regression model. This approach aimed to account for the specific characteristics and dynamics associated with each province. A two-year period was used for each data point to attempt to capture present lingering frustrations caused by a continuous state of poor socio-economic conditions and dissatisfaction.

The study established a positive linear relationship between unrest intensity and poor service delivery. This finding is supported by Frustration-Aggression Theory and studies conducted by Kali (2023), Nleya (2011), and Botes (2018). In areas where service delivery is worse, there is thus a higher probability of witnessing elevated levels of protest intensity.

Some unexpected results were observed, such as the lack of significance between protest intensity and income level or political dissatisfaction. These factors were found to be important determinants of unrest in the studies conducted by Kali (2023), Nleya (2011), Ndasana et al. (2022), and Botes (2018). It is therefore suggested that further investigation with different questions or measures (indices with more complex weightings and questions more closely associated with income level and political dissatisfaction) could provide deeper insights.

By identifying and quantifying the factors that significantly contribute to unrest intensity, a better understanding of the complex dynamics between unrest and its predictors is developed. This will allow policymakers and stakeholders to more effectively reduce the intensity of socio-political unrest. The established framework that utilises GDELT Project data can effectively be applied in any country due to the global coverage provided by the project.

Future Work

An investigation into more complex indices with associated weights could prove very fruitful. Making use of additional data more closely related to income levels and political dissatisfaction could also improve model results and account for the observed unexpected, albeit insignificant, results. Adding layers of categorisation to the GDELT Project protest data (violent protests, service delivery-related protests, etc.) and conducting an analysis utilising the established framework could also lead to interesting results, potentially shedding more light on the complex dynamics of protest intensity and its predictors.

Appendix:

Definitions and Formulas

GDELT Project – A real-time network diagram and database of global human society for open research.

Afrobarometer – A pan-African survey research network that conducts public attitude surveys on democracy, governance, the economy, and society.

Haversine Formula – A formula that determines the great-circle distance between two points on a sphere given their longitudes and latitudes.

$$\theta = \frac{d}{r}$$

$$\text{hav}(\theta) = \text{hav}(\varphi_2 - \varphi_1) + \cos(\varphi_1) \cos(\varphi_2) \text{hav}(\lambda_2 - \lambda_1)$$

Likert Scales – A psychometric scale which is commonly used in research questionnaires that contain scaling responses.

Ordinal Encoding – A method used to convert ordinal categorical data into ordered numerical data.

Binary encoding – A method used to convert categorical data with variables into numerical data by assigning 1s and 0s.

One-Hot Encoding – A technique that can be used to represent categorical variables as numerical values in machine learning by taking a variable with multiple unique entries and breaking up the variable and assigning 1s and 0s to each unique entry.

Afrobarometer Survey Questions

Income Questions

EA-FAC. Are the following facilities present in the primary sampling unit / enumeration area or in easy walking distance?	Yes	No	Can't Determine
B. School (private or public or both)	0	1	9
C. Police station	0	1	9
D. Health clinic (private or public or both)	0	1	9

Let's discuss economic conditions						
4. In general, how would you describe:						
	Very good	Fairly good	Neither good nor bad	Fairly bad	Very bad	Don't know
B. Your own present living conditions?	1	2	3	4	5	9

8. Over the past year, how often, if ever, have you or anyone in your family:						
	Never	Just once or twice	Several times	Many times	Always	Don't know
A. Gone without enough food to eat?	1	2	3	4	5	9
B. Gone without enough clean water for home use?	1	2	3	4	5	9
C. Gone without medicines or medical treatment?	1	2	3	4	5	9
D. Gone without enough fuel to cook your food?	1	2	3	4	5	9
E. Gone without a cash income?	1	2	3	4	5	9

Service Delivery Questions

EA-ROAD-C Thinking of the journey here:	Impassable	Very poor	Poor	Fair	Good	Very good
What was the condition of the road in the last 5 km before reaching the start point of the PSU / EA?	6	5	4	3	2	1

49. Now I would like to talk to you about experiences that some people have in accessing certain essential government services

B. How easy or difficult was it to obtain the services you needed from teachers or school officials?	No contact	Very easy	Easy	Difficult	Very difficult	Don't know
E. How easy or difficult was it to obtain the medical care you needed?	9	1	2	3	4	9
P. How easy or difficult was it to obtain the assistance you needed?	9	1	2	3	4	9

Political Satisfaction Questions

3. Let's start with your general view about the current direction of our country. Some people might think the country is going in the wrong direction. Others may feel it is going in the right direction. So let me ask YOU about the overall direction of the country: Would you say that the country is going in the wrong direction or going in the right direction?	
Going in the wrong direction	1
Going in the right direction	0
Don't know	9

Let's discuss economic conditions						
4. In general, how would you describe:						
	Very good	Fairly good	Neither good nor bad	Fairly bad	Very bad	Don't know
A. The present economic condition of this country?	1	2	3	4	5	9

36. Overall, how satisfied are you with the way democracy works in South Africa?	
Very satisfied?	1
Fairly satisfied?	2
Don't know	9
Not very satisfied?	3
Not at all satisfied?	4
South Africa is not a democracy	5
Don't know	9

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